CS246: Database Management Systems Lab

SQL Triggers and Exceptions

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Overview

Triggers	Performs operations based on actions
Exceptions	Declare & Use

Triggers - Introduction - 1

Event-condition-action (ECA) rule

- When an event occurs, test the specified condition.
- If the condition is true, execute the given actition

Example

Event	When `Table1` gets `updated` on
Condition	If new value is $\geq$ old value
Action	insert a row into `Table2`

Triggers - Introduction - 2

```
CREATE TRIGGER first_trigger
AFTER UPDATE ON Table1                                          Event
FOR EACH ROW                                                    condition
BEGIN
  IF (NEW.price > OLD.price)
   THEN
    INSERT into Table2(np, op) values (NEW.price, OLD.price);   action
  END IF;
END
```

Possible events

- On inserting a row into a table on which trigger is specified
- On deleting a row from a table on which trigger is specified
- On Updating a row in a table on which trigger is specified

Triggers - Introduction - 3

Possible events

- On inserting a row into a table on which trigger is specified
- On deleting a row from a table on which trigger is specified
- On Updating a row in a table on which trigger is specified

Granularity

- For each row
- For each statement

Triggers - Introduction - 3

Possible events

- On inserting a row into a table on which trigger is specified
- On deleting a row from a table on which trigger is specified
- On Updating a row in a table on which trigger is specified

Granularity

- For each row
- For each statement

Timing

- Before the event
- After the event

Triggers - Introduction - 4

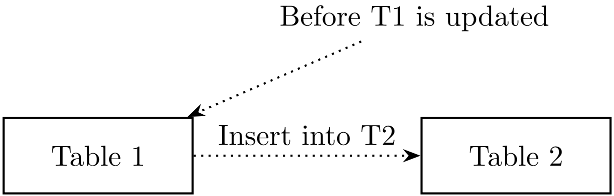
Statement level triggers

Not implemented in MySQL

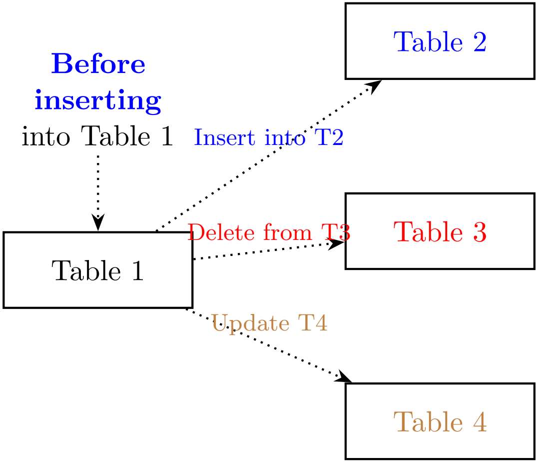
Row level triggers

Trigger gets fired for each affected row of the table

Trigger - Exmple 01




```
DELIMITER |
CREATE TRIGGER before_products_update
  BEFORE UPDATE ON Table1
  FOR EACH ROW
BEGIN
    IF OLD.msrp <> NEW.msrp THEN
      INSERT INTO Table2(product_code,price)
    VALUES(old.productCode,old.msrp);
  END IF;
END |
DELIMITER ;
```




```
DELIMITER |

CREATE TRIGGER testref BEFORE INSERT ON Table1
  FOR EACH ROW
  BEGIN
    INSERT INTO Table2 SET a2 = NEW.a1;
    DELETE FROM Table3 WHERE a3 = NEW.a1;
    UPDATE Table4 SET b4 = b4 + 1 WHERE a4 = NEW.a
  END;|
DELIMITER;
```

Trigger - Example 02

Table 1
1

Table 2

Table 3
1
2
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Table 4
(1, 0)
(2, 0)
(3, 0)
(4, 0)
(5, 0)
(6, 0)
(7, 0)
(8, 0)
(9, 0)
(10, 0)

Table 4

Trigger - Example 02

Table 1
1

Table 2
1

Table 3
1
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8
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Table 4
(1, 0)
(2, 0)
(3, 0)
(4, 0)
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Table 4

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
(1, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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(10, 0)

Table 4
(1, 1)

Trigger - Example 02

Table 1
1
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Table 2
1

Table 3
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Table 4
~~(1, 0)~~
(2, 0)
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Table 4
(1, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
(1, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
(1, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
(1, 1)
(2, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
~~1~~
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Table 4
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Table 4
(1, 1)
(2, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 4
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Table 4
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Trigger - Example 02

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Table 4
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Table 4
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Trigger - Example 02

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Table 4
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Table 4
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Trigger - Example 02

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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 4
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Table 4
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Trigger - Example 02

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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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(4, 1)
(5, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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(2, 1)
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Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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Table 4
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(5, 1)
(6, 1)

Trigger - Example 02

Table 1
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Table 2
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Table 3
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Table 4
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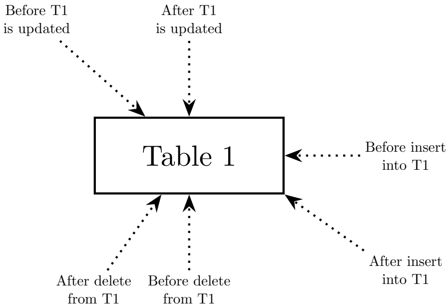
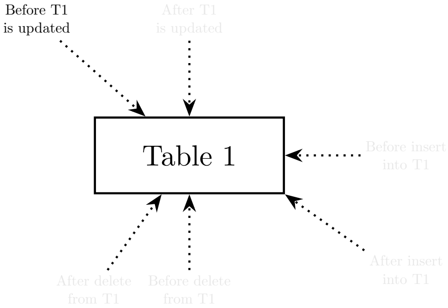
Table 4
(1, 1)
(2, 1)
(3, 1)
(4, 1)
(5, 1)
(6, 1)

Trigger - multiple triggers on single table



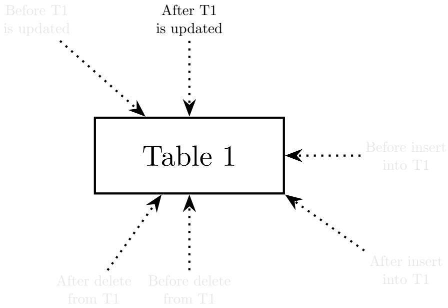

```
DELIMITER |
CREATE TRIGGER trigger_name
   {BEFORE | AFTER} {INSERT | UPDATE | DELETE}
   ON table
   FOR EACH ROW
   {FOLLOWS | PRECEEDS}
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 2



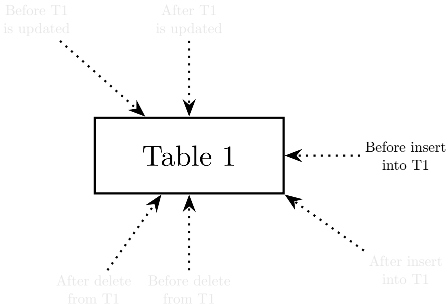

```
DELIMITER |
CREATE TRIGGER trigger_2
  BEFORE UPDATE
  ON Table1
  FOR EACH ROW
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 3



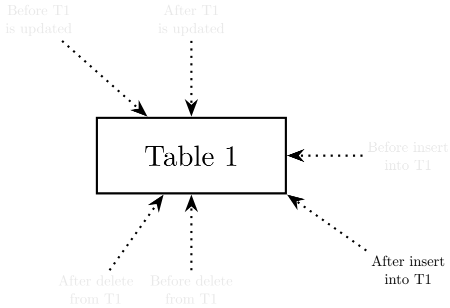

```
DELIMITER |
CREATE TRIGGER trigger_3
   AFTER UPDATE
   ON Table1
   FOR EACH ROW
   FOLLOWS trigger_2 | PRECEEDS
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 4



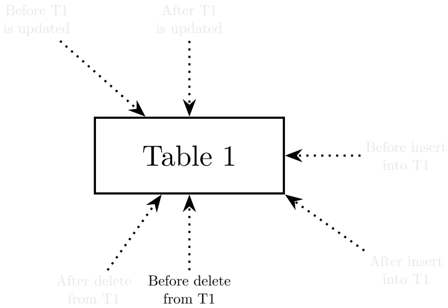

```
DELIMITER |
CREATE TRIGGER trigger_4
  BEFORE INSERT
  ON Table1
  FOR EACH ROW
  PRECEEDS trigger_3
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 5



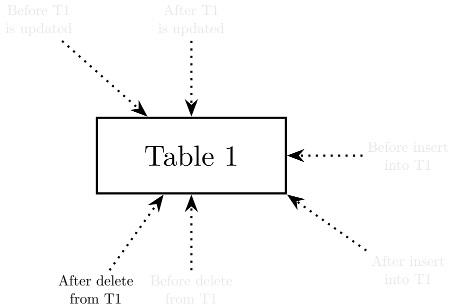

```
DELIMITER |
CREATE TRIGGER trigger_5
   AFTER INSERT
   ON Table1
   FOR EACH ROW
   FOLLOWS trigger_4
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 6



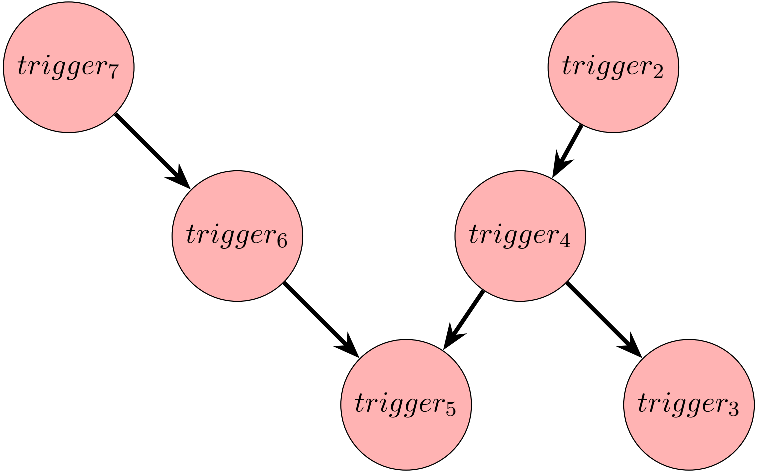

```
DELIMITER |
CREATE TRIGGER trigger_6
   BEFORE DELETE
   ON Table1
   FOR EACH ROW
   PRECEEDS trigger_5
BEGIN
END |
DELIMITER ;
```

Trigger - multiple triggers on single table - 7

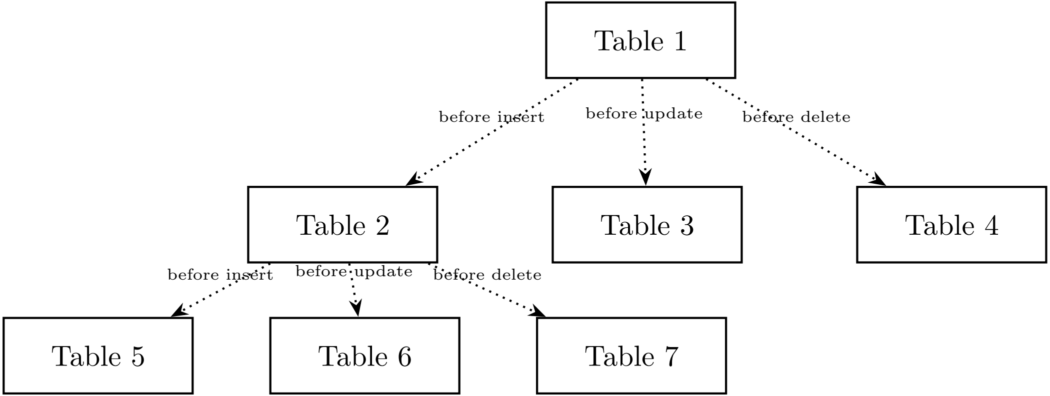



```
DELIMITER |
CREATE TRIGGER trigger_7
   AFTER DELETE
   ON Table1
   FOR EACH ROW
   PRECEEDS trigger_6
BEGIN
END |
DELIMITER ;
```

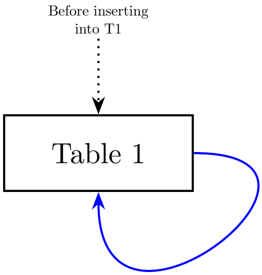
Trigger - multiple triggers on single table - preceedence graph



Trigger - nested triggers



Triggers & cycles? - direct cycle



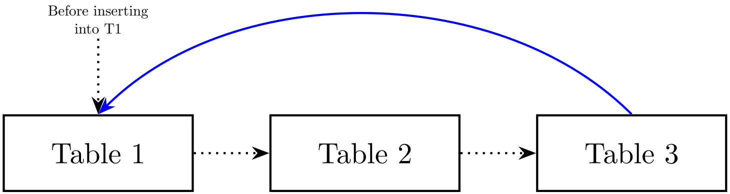

```
DELIMITER |
CREATE TRIGGER trigger_8
  BEFORE INSERT
  ON Table1
  FOR EACH ROW
  BEGIN
   INSERT INTO Table1(c1, c2) VALUES (1, 2);
  END |
DELIMITER ;
```

Triggers & cycles? - indirect cycle

```
DELIMITER |
CREATE TRIGGER trigger_3
  BEFORE INSERT
  ON Table3
  FOR EACH ROW
  BEGIN
   INSERT INTO Table1(c1, c2) VALUES (1, 2);
  END |
DELIMITER ;
```

```
DELIMITER |
CREATE TRIGGER trigger_2
  BEFORE INSERT
  ON Table2
  FOR EACH ROW
  BEGIN
   INSERT INTO Table3(c1, c2) VALUES (1, 2);
  END |
DELIMITER ;
```

```
DELIMITER |
CREATE TRIGGER trigger_1
  BEFORE INSERT
  ON Table1
  FOR EACH ROW
  BEGIN
   INSERT INTO Table2(c1, c2) VALUES (1, 2);
  END |
DELIMITER ;
```



Triggers - notation

Sailor

OLD.sid	OLD.sname	OLD.rating	OLD.age
22	Dustin	7	45.0
29	Brutus	1	33.0
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35.0
64	Horatio	7	35.0
71	Zorba	10	16.0
74	Horatio	9	35.0
85	Art	3	25.5
95	Bob	3	63.5
105	Robin	9	27.60

New record

NEW.sid	NEW.sname	NEW.rating	New.age
105	Robin	9	27.60

```
INSERT INTO Sailor(sid, sname, rating, age)
VALUES(105, 'Robin', 9, 27.60);
```

Inside TRIGGER

- IF NEW.sid == 105 refers to new row sid to be inserted into Sailor table
- IF OLD.sid == 32 refers to old row sid in the Sailors table

Exceptions - Example 01

```
delimiter $$
CREATE PROCEDURE insert_user(IN name VARCHAR(50), IN email VARCHAR(100))
BEGIN
    DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
    BEGIN
        SELECT 'Duplicate entry error' AS ErrorMessage;
    END;

    INSERT INTO my_users (name, email) VALUES (name, email);
END $$

DELIMITER ;
```

Exceptions - Example 01

CALL insert_user('user1', 'user1@example.com');
CALL insert_user('user1', 'user1@example.com'); – Caught by exception

Exceptions - Example 02

```
DELIMITER $$

CREATE PROCEDURE test_missing_table()
BEGIN
    DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
    BEGIN
        SELECT 'Table does not exist!' AS ErrorMessage;
    END;

    -- Assume MyTable is not present in the database
    SELECT * FROM MyTable;
END $$
DELIMITER ;
```

$\boxed{\text{CALL test\_missing\_table() – Caught by exception}}$

Exceptions - Example 03

```
DELIMITER $$

CREATE PROCEDURE divide_numbers(IN num1 INT, IN num2 INT)
BEGIN
    DECLARE CONTINUE HANDLER FOR SQLEXCEPTION
    BEGIN
        SELECT 'Division by zero error!' AS ErrorMessage;
    END;

    SELECT num1 / num2 AS result;
END $$
DELIMITER ;
```

Exceptions - Example 03

CALL divide_numbers(10, 0); – Caught by Exception

Exceptions - Example 04

```
DELIMITER $$

CREATE PROCEDURE two_exceptions()
BEGIN
    DECLARE EXIT HANDLER FOR SQLSTATE '42S02' -- Table not found
    BEGIN
        SELECT 'Table not found error!' AS ErrorMessage;
    END;

    DECLARE EXIT HANDLER FOR SQLSTATE '23000' -- Integrity constraint violation
    BEGIN
        SELECT 'Integrity constraint violation!' AS ErrorMessage;
    END;

    -- Insert into a non-existent table
    INSERT INTO MyTable VALUES (1, 'Test');
END $$
DELIMITER ;
```